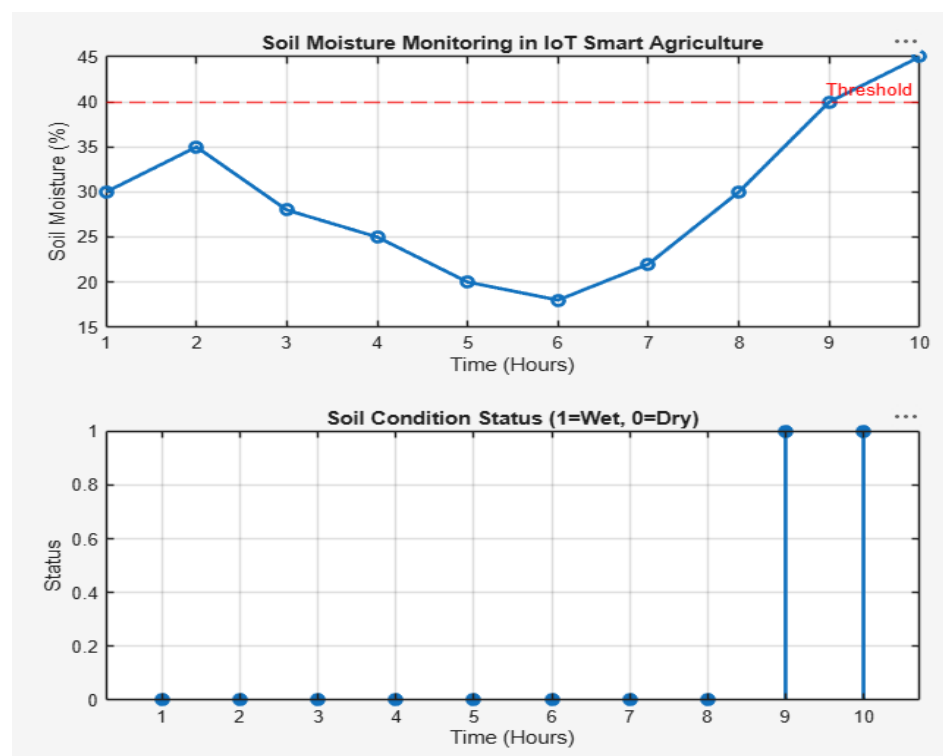


Ex.No.11: Implementation of IoT-based Smart Agriculture using MATLAB

Objective 1: Matlab Code and Output

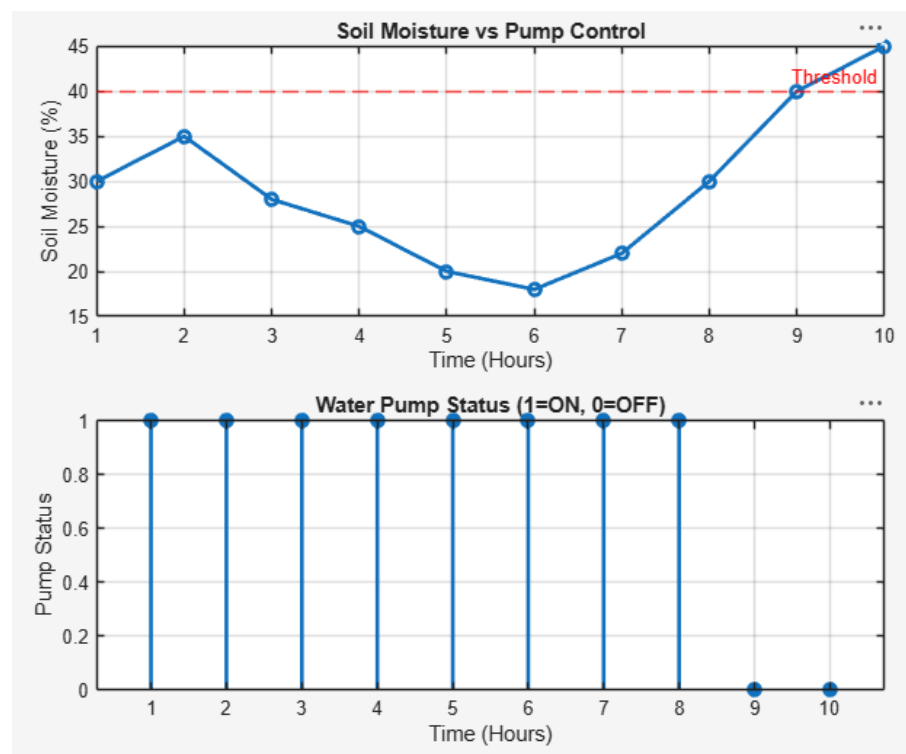
/MATLAB Drive/slotb11.m

```
1  clc; clear; close all;
2  % Simulated soil moisture data (%)
3  time = 1:10;
4  soil_moisture = [30 35 28 25 20 18 22 30 40 45];
5  % Threshold for irrigation
6  threshold = 40;
7  figure
8  % ----- Graph 1: Soil Moisture Variation -----
9  subplot(2,1,1)
10 plot(time, soil_moisture, '-o', 'LineWidth', 2)
11 title('Soil Moisture Monitoring in IoT Smart Agriculture')
12 xlabel('Time (Hours)')
13 ylabel('Soil Moisture (%)')
14 grid on
15 hold on
16 yline(threshold, 'r--', 'Threshold')
17 % ----- Graph 2: Moisture Status (Dry/Wet Level) -----
18 status = soil_moisture >= threshold;
19 subplot(2,1,2)
20 stem(time, status, 'filled', 'LineWidth', 2)
21 title('Soil Condition Status (1=Wet, 0=Dry)')
22 xlabel('Time (Hours)')
23 ylabel('Status')
24 grid on
```



Objective 2: Matlab Code and Output

```
26 % Simulated soil moisture (%)
27 time = 1:10;
28 soil_moisture = [30 35 28 25 20 18 22 30 40 45];
29 % Threshold level
30 threshold = 40;
31 % Pump control logic (1 = ON, 0 = OFF)
32 pump_status = soil_moisture < threshold;
33 figure
34 % ----- Graph 1: Soil Moisture vs Pump Status -----
35 subplot(2,1,1)
36 plot(time, soil_moisture, '-o','LineWidth',2)
37 title('Soil Moisture vs Pump Control')
38 xlabel('Time (Hours)')
39 ylabel('Soil Moisture (%)')
40 grid on
41 hold on
42 yline(threshold,'r--','Threshold')
43 % ----- Graph 2: Pump Status (ON/OFF) -----
44 subplot(2,1,2)
45 stem(time, pump_status, 'filled','LineWidth',2)
46 title('Water Pump Status (1=ON, 0=OFF)')
47 xlabel('Time (Hours)')
48 ylabel('Pump Status')
49 grid on
```



Objective 3: Matlab Code and Output

```
51 % Simulated soil moisture data (%)
52 time = 1:10;
53 soil_moisture = [30 35 28 25 20 18 22 30 40 45];
54 % Threshold level
55 threshold = 40;
56 % Irrigation efficiency calculation (%)
57 efficiency = (soil_moisture ./ threshold) * 100;
58 efficiency(efficiency > 100) = 100; % limit to 100%
59 figure
60 % ----- Graph 1: Soil Moisture vs Efficiency -----
61 subplot(2,1,1)
62 plot(time, soil_moisture, '-o','LineWidth',2)
63 title('Soil Moisture Variation')
64 xlabel('Time (Hours)')
65 ylabel('Soil Moisture (%)')
66 grid on
67 hold on
68 yline(threshold,'r--','Threshold')
69 % ----- Graph 2: Irrigation Efficiency -----
70 subplot(2,1,2)
71 plot(time, efficiency, '-s','LineWidth',2)
72 title('Irrigation Efficiency (%) in Smart Agriculture')
73 xlabel('Time (Hours)')
74 ylabel('Efficiency (%)')
75 grid on
```

